Application of the Equality Principle to Social Media Algorithms

SCX

SCX and Social Media Information Potential Energy Surface:
Recommendation Algorithms as Information Potential Isolators
and the Convergence Condition of Cross-Coordinate-System Exposure

SCX

202671

Abstract

Contemporary social media platforms rely on recommendation algorithms that maximize user engagement, but these algorithms simultaneously act as **Information Potential Isolators**. This paper applies the Equality Principle ($\sum g_m = 0$) from the SCX framework to the information propagation dynamics of social media, establishing a rigorous mathematical formalization of the Information Potential Energy Surface (IPES). We prove that: **(1)** recommendation algorithms, by showing users content matching their existing coordinate systems, drive the cross-group coupling strength $g_{\text{cross}} \to 0$, locking different user groups into their respective information potential wells — this is the potential-energy explanation of the **Filter Bubble**; **(2)** when $g_{\text{cross}} = 0$, user groups in different coordinate systems oscillate permanently without intersecting, a direct consequence of Corollary 8 — **Polarization** is the macroscopic manifestation of permanent oscillation; **(3)** the isolation law of the IPES yields a new standard for algorithmic auditing: the audit item g_{cross} measures the intensity of cross-coordinate-system exposure; **(4)** the solution follows from the core requirement of the Equality Principle — Forced Mixed Information Streams, i.e., artificially increasing g_{cross} to drive the system toward mutual understanding. This paper provides a complete theorem system, formal proofs, numerical examples, and honest critique markings.

/Keywords

Information Potential Energy Surface; Recommendation Algorithm; Filter Bubble; Polarization; Cross-Coordinate-System Exposure; Algorithmic Audit; Equality Principle

Contents

1

2 Introduction: The Potential Energy Crisis in the Age of Social Media

2.1

202650 ——Recommendation Algorithm—— Engagement

Information Potential Wells

In the year 2026, over 5 billion people worldwide receive information through social media, and the core driving force of these platforms — the recommendation algorithm — is optimized to maximize user engagement. Superficially, this is a natural extension of business logic: platforms want users to stay longer, click more, and interact more frequently. However, from the perspective of information dynamics, this optimization process is systematically partitioning an originally connected information space into mutually impermeable **Information Potential Wells**.

Definition 2.1 (Information Potential Energy Surface, IPES)**.** $\mathcal{X}$ $\mathcal{Y}$

$$\mathcal{V} : \mathcal{X} \times \mathcal{Y} \to \mathbb{R},$$

$\mathcal{V}(x, y)$ y x

$G \subset \mathcal{X}$ $\mathcal{V}$ $\mathcal{V}$ $\mathcal{V}$

Let $\mathcal{X}$ be the user population space and $\mathcal{Y}$ be the information content space. The Information Potential Energy Surface is a scalar field $\mathcal{V} : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$, where $\mathcal{V}(x, y)$ represents the cognitive potential energy of information item y for user x. Lower potential energy means the user finds the information easier to accept (lower cognitive resistance). Each user group $G \subset \mathcal{X}$ forms a ***potential well*** *on $\mathcal{V}$: users tend to consume and share information that minimizes $\mathcal{V}$, and reject information that raises $\mathcal{V}$.*

Definition 2.2 (Coordinate System)**.** $\mathcal{C}_G$ $\{\mathbf{e}_1, \ldots, \mathbf{e}_d\}$ *d-* A B

$$\Delta(\mathcal{C}_A, \mathcal{C}_B) = \inf_{\mathbf{R} \in O(d)} \|\mathcal{C}_A - \mathbf{R}\mathcal{C}_B\|_F,$$

$O(d)$ $\| \cdot \|_F$ *Frobenius*

”

”

A user group's ***Information Coordinate System*** *$\mathcal{C}_G$ is a d-dimensional subspace spanned by orthogonal basis vectors $\{\mathbf{e}_1, \ldots, \mathbf{e}_d\}$, through which the group interprets world events. The* ***coordinate system distance*** *between two groups A and B is defined as $\Delta(\mathcal{C}_A, \mathcal{C}_B)$, measuring the extent to which different groups have divergent "interpretive frameworks" for the same events.*

2.2 =

2.3 Core Insight: Recommendation Algorithm = Information Potential Isolator

$$=$$

$$\text{Recommendation Algorithm} = \text{Information Potential Isolator}$$

$$\Rightarrow g_{\text{cross}} \to 0$$

$$\Rightarrow \ 8$$

$$\Rightarrow g_{\text{cross}}$$

Equality Principle $\sum_m g_m = 0$——

g_{cross} adiabatic wall——

This insight comes from the direct application of the Equality Principle ($\sum_m g_m = 0$) to information propagation scenarios. The Equality Principle requires that the sum of all cross-group interaction coupling strengths vanish — this does not mean interactions are absent, but that the **net potential energy flow vanishes at equilibrium**. The problem with recommendation algorithms is that they do more than merely let the net flow vanish — they **systematically prevent any cross-group interaction from occurring**, driving g_{cross} itself to zero. This is equivalent to inserting an adiabatic wall between potential wells — each group oscillates eternally in its own well, never contacting the coordinate framework of other groups.

2.4

2.5 Main Contributions

1 —— **2 .**

g_{cross}

Contribution 1 — Formalization of IPES (Section 2). We establish the potential energy surface model of social media information propagation, defining core concepts including information potential, coordinate system, potential well, and coupling strength g_{cross}, and derive the fundamental dynamical equations of the IPES.

2 —— **3** . g_{cross} $g_{\text{cross}} \to 0$

Contribution 2 — Isolation Theorem of Recommendation Algorithms (Section 3). We prove that the optimization objective of recommendation algorithms (maximizing engagement) is equivalent to minimizing cross-coordinate-system exposure g_{cross}, and give the rigorous conditions for permanent oscillation when $g_{\text{cross}} \to 0$.

3 —— **4** . g_{cross} g_{cross}

Contribution 3 — Algorithmic Audit Standard (Section 4). We propose an algorithmic audit framework based on g_{cross} and prove its measurability and lower bound.

4 —— **5** . g_{cross}

Contribution 4 — Convergence Guarantee of Forced Mixed Streams (Section 5). We prove that artificially increasing g_{cross} drives the system from permanent oscillation to convergence toward understanding consensus, and give the convergence rate.

3

4 Mathematical Formalization of the Information Potential Energy Surface

4.1

4.2 Basic Setup and Notation

K $\mathcal{X} = \{G_1, \ldots, G_K\}$ G_k N_k $\mathcal{C}_k$ $\mathcal{Y} \subset \mathbb{R}^D$ $t = 0, 1, 2, \ldots$

Consider a society $\mathcal{X} = \{G_1, \ldots, G_K\}$ with K user groups, each of size N_k, possessing its own information coordinate system $\mathcal{C}_k$. The information content space $\mathcal{Y} \subset \mathbb{R}^D$ is the embedding representation of all possible information items (posts, articles, videos, etc.). Time is discrete, in units of algorithm recommendation cycles: $t = 0, 1, 2, \ldots$

Definition 4.1 (Group Information State)**.** G_k t

$$\mu_k^{(t)} \in \mathcal{P}(\mathcal{Y}),$$

$\pi_k : \mathcal{P}(\mathcal{Y}) \to \mathcal{P}(\mathcal{Y})$

$$\mu_k^{(t+1)} = \pi_k(\mu_k^{(t)}).$$

*The **information state** of group G_k at time t is a probability measure $\mu_k^{(t)} \in \mathcal{P}(\mathcal{Y})$ representing the distribution of information content currently consumed by that group. Its evolution is driven by a recommendation policy $\pi_k : \mathcal{P}(\mathcal{Y}) \to \mathcal{P}(\mathcal{Y})$.*

Definition 4.2 (Information Potential Energy)**.** $\mu \in \mathcal{P}(\mathcal{Y})$ $\mathcal{C}$

$$\mathcal{V}(\mu; \mathcal{C}) = \int_{\mathcal{Y}} \|_{\mathcal{C}^\perp}(y)\|^2 \, d\mu(y), \tag{1}$$

$\mathcal{C}^\perp$

Given information state $\mu \in \mathcal{P}(\mathcal{Y})$ and coordinate system $\mathcal{C}$, the group's information potential energy is defined as in (1), where $_{\mathcal{C}^\perp}$ is the projection onto the orthogonal complement of the coordinate system. Intuition: the component of an information item orthogonal to the group's coordinate system contributes cognitive resistance (potential energy). Components within the coordinate system contribute zero potential energy (naturally accepted).

Definition 4.3 (Alignment)**.** G_k G_j

$$A_{kj}(\mu_k) = 1 - \frac{\mathcal{V}(\mu_k; \mathcal{C}_j)}{\max_\mu \mathcal{V}(\mu; \mathcal{C}_j)}. \tag{2}$$

$A_{kk} = 1 A_{kj} \in [0,1]$ k j

*The **alignment** between G_k's information state and G_j's coordinate system is $A_{kj} = 1$ for self-alignment; $A_{kj} \in [0,1]$ measures the compatibility between the information consumed by group k and the interpretive framework of group j.*

4.3

4.4 Dynamics of the Information Potential Energy Surface

Assumption 4.4. π_k

$$\mathcal{V}(\pi_k(\mu); \mathcal{C}_k) \leq \mathcal{V}(\mu; \mathcal{C}_k), \quad \forall \mu \in \mathcal{P}(\mathcal{Y}).$$

Users (and the recommendation algorithms serving them) tend to minimize potential energy: consume information aligned with their own coordinate system, avoid information requiring cognitive restructuring. Formally, the recommendation policy π_k satisfies the above inequality for all μ.

Definition 4.5 (Cross-Group Coupling Strength)**.** G_k G_j

$$g_{cross}{}_{kj}^{(t)} = \frac{1}{N_k N_j} \sum_{x \in G_k} \sum_{y \in G_j} \exp -\alpha \cdot \Delta(\mathcal{C}_k, \mathcal{C}_j) \cdot \mathbb{1}\{x \ y \ t\}, \tag{3}$$

$\alpha > 0$ $g_{crosskj}$ k j

*The **cross-group coupling strength** between groups G_k and G_j is defined as in (3), where $\alpha > 0$ is the decay coefficient of coordinate system distance. $g_{crosskj}$ measures the effective intensity with which members of group k are exposed to information in the coordinate system of group j.*

Definition 4.6 (System Coupling Matrix)**.** $K \times K$

$$\mathbf{G}^{(t)} = [g_{cross}{}_{kj}^{(t)}]_{k,j=1}^{K},$$

$g_{crosskk} = 1$ $g_{crosskj} \in [0,1]$

Define the $K \times K$ coupling matrix $\mathbf{G}^{(t)} = [g_{cross}{}_{kj}^{(t)}]_{k,j=1}^{K}$, with diagonal $g_{crosskk} = 1$ (intra-group coupling is always maximal), off-diagonal $g_{crosskj} \in [0,1]$ measuring cross-group exposure.

Definition 4.7 (Potential Well Depth)**.** G_k

$$D_k^{(t)} = \mathcal{V}(\mu_k^{(t)}; \mathcal{C}_k), \tag{4}$$

$D_k^{(t)}$

*The **depth** of group G_k in its own information potential well is defined as in (4), i.e., the potential energy of the group's information state under its own coordinate system. Due to the potential energy minimization drive, $D_k^{(t)}$ is non-increasing in time.*

4.5

4.6 Total Potential Energy and the Equality Condition

Definition 4.8 (Total Social Information Potential Energy)**.** t

$$\mathcal{V}_{total}^{(t)} = \sum_{k=1}^{K} \mathcal{V}(\mu_k^{(t)}; \mathcal{C}_k). \tag{5}$$

Proposition 4.9 (Monotonicity of Total PE)**.**

$$\mathcal{V}_{total}^{(t+1)} \leq \mathcal{V}_{total}^{(t)}, \quad \forall t \geq 0.$$

$$\mathcal{V}(\mu_k^{(t)}; \mathcal{C}_k) = 0, \quad \forall k.$$

Under pure recommendation drive (no cross-group intervention), the total information potential energy decreases monotonically, with equality iff all groups have reached the bottom of their potential wells.

Proof. 1 π_k $\mathcal{V}(\mu_k^{(t+1)}; \mathcal{C}_k) \leq \mathcal{V}(\mu_k^{(t)}; \mathcal{C}_k)$ k □

—— = =

This proposition appears innocuous — decreasing potential energy seems to mean users are becoming more comfortable. However, this is precisely the mathematical expression of the crisis: **total potential energy minimization = each group sinking deep into its own potential well = cross-group isolation maximization**.

Honest Critique [!] **1**. . 4.9 ——

$g_{\text{cross}kj}$

Comfort is Isolation. Proposition 4.9 reveals a grim mathematical fact: recommendation algorithms drive each group's potential energy to zero under its own coordinate system — meaning users see information that perfectly fits their existing framework, with zero cognitive friction. But simultaneously, the coupling strength $g_{\text{cross}kj}$ between different groups tends to zero. **Global potential energy minimization and global connectivity maximization are mutually incompatible.** Choosing comfort (low PE) means choosing isolation (low connectivity).

5

6 The Information Potential Isolation Theorem of Recommendation Algorithms

6.1

6.2 Recommendation Algorithm as Potential Energy Minimizer

Transformer

Modern recommendation algorithms (collaborative filtering, deep neural network recommenders, Transformer-based sequential recommenders, etc.) can be uniformly understood as solving the following optimization problem:

Definition 6.1 (Objective of Recommendation Algorithm)**.** $x \in G_k$ $y^* \in \mathcal{Y}$

$$y^* = \arg\max_{y \in \mathcal{Y}} \underbrace{s(x, y)} - \underbrace{\beta \cdot \mathcal{V}(y; \mathcal{C}_k)}, \tag{6}$$

$s(x, y)$ $\beta \geq 0$

Given a user $x \in G_k$, the recommendation algorithm selects an information item $y^* \in \mathcal{Y}$ to maximize the expression in (6)*, where $s(x, y)$ is the predicted engagement score (click-through rate, dwell time, share probability, etc.), and $\beta \geq 0$ is the potential energy penalty weight.*

Remark 6.2. (6) $\mathcal{V}(y;\mathcal{C}_k)$

—— Selection Bias

The potential energy penalty $\mathcal{V}(y;\mathcal{C}_k)$ in (6) is typically not explicitly designed but implicitly embedded in the training data from which the algorithm learns. Since the algorithm learns from historical interaction data, and users in the historical data already tend to consume low-PE content — this is the potential-energy form of **Selection Bias**.

Theorem 6.3 (Potential Isolation Theorem of Recommendation)**.** (6) $\beta > 0$ $G_k \neq G_j$

$$g_{cross_{kj}}^{(t)} \leq g_{cross_{kj}}^{(0)} \cdot \exp -\beta \cdot \Delta(\mathcal{C}_k, \mathcal{C}_j) \cdot t. \tag{7}$$

$t \to \infty$

$$\lim_{t\to\infty} g_{cross_{kj}}^{(t)} = 0, \quad \forall k \neq j.$$

Let the recommendation algorithm optimize (6) *with* $\beta > 0$*. Then for any two distinct groups* $G_k \neq G_j$*, the cross-group coupling strength satisfies* (7)*. In particular, as* $t \to \infty$*,* $g_{cross_{kj}}^{(t)} \to 0$ *for all* $k \neq j$*.*

Proof. k t $\mathcal{Y}_k^{(t)}$ $\beta \cdot \mathcal{V}(y;\mathcal{C}_k)$ $\mathcal{C}_k$ $\|_{\mathcal{C}_k^\perp}(y)\|$ y j $y \in \mathcal{C}_j$

$$\mathcal{V}(y;\mathcal{C}_k) = \|_{\mathcal{C}_k^\perp}(y)\|^2 \geq \sin^2(\theta_{kj}) \cdot \|y\|^2,$$

θ_{kj} $\mathcal{C}_k$ $\mathcal{C}_j$ principal angle $\sin(\theta_{kj}) \geq c \cdot \Delta(\mathcal{C}_k, \mathcal{C}_j)$ $c > 0$ j $\exp(-\beta \cdot \Delta(\mathcal{C}_k, \mathcal{C}_j))$ t (7)

Consider the set $\mathcal{Y}_k^{(t)}$ of information items consumed by group k at time t. The PE penalty $\beta \cdot \mathcal{V}(y;\mathcal{C}_k)$ biases the algorithm toward content within $\mathcal{C}_k$, i.e., y with small $\|_{\mathcal{C}_k^\perp}(y)\|$. For content $y \in \mathcal{C}_j$ in group j's coordinate system, we have $\mathcal{V}(y;\mathcal{C}_k) \geq \sin^2(\theta_{kj}) \cdot \|y\|^2$, where θ_{kj} is the principal angle between the two coordinate systems. By definition of coordinate system distance, $\sin(\theta_{kj}) \geq c \cdot \Delta(\mathcal{C}_k, \mathcal{C}_j)$ for some absolute constant $c > 0$. Therefore, in each recommendation cycle, the probability that information from group j's coordinate system is selected decays by factor $\exp(-\beta \cdot \Delta(\mathcal{C}_k, \mathcal{C}_j))$. Accumulating over t steps yields the exponential decay bound (7). □

Corollary 6.4 (Potential-Energy Reformulation of Filter Bubble)**.** *6.3* $g_{crosskj} = 0$ $k \neq j$ *Complete Potential Isolation*

Theorem 6.3 provides a precise potential-energy reformulation of the filter bubble phenomenon: the filter bubble is not an accidental byproduct but an ***inevitable consequence of the recommendation algorithm's gradient descent on the potential energy surface****. When* $g_{crosskj} = 0$ *for all* $k \neq j$*, the system is in a state of* ***Complete Potential Isolation****: each group is fully enclosed in its own potential well, and any information from other coordinate systems is reflected by the potential barrier.*

Honest Critique **[!] 2**. . (7) $\beta \cdot \Delta(\mathcal{C}_k, \mathcal{C}_j)$ β $\Delta(\mathcal{C}_k, \mathcal{C}_j)$ Δ

The Speed of Exponential Decay. The bound (7) shows that isolation occurs at exponential speed. Both factors in the decay rate, β (fitting strength of the algorithm to historical data) and $\Delta(\mathcal{C}_k, \mathcal{C}_j)$ (granularity of content classification and user clustering), are products of platform design. In practice, Δ is typically large from the start (because platforms already finely partition users for ad targeting), meaning that **isolation begins from the very first day a user registers**.

6.3

6.4 The Permanent Oscillation Theorem

Once the system reaches complete potential isolation, how do the information states of different groups evolve? The following theorem provides the answer.

Theorem 6.5 (Permanent Oscillation in IPE Wells)**.** $g_{crosskj} = 0$ $k \neq j$ π_k *ergodic* $\mu_k^{(t)}$ $\mathcal{W}_k = \{\mu : \mathcal{V}(\mu; \mathcal{C}_k) = 0\}$

$$(\mu_k^{(t)}) \cap (\mu_j^{(t')}) = \varnothing, \quad \forall k \neq j, \forall t, t' \geq T_{iso},$$

T_{iso}

Assume the system is in complete potential isolation ($g_{crosskj} = 0$ for all $k \neq j$), and each group's recommendation policy π_k is ergodic within its potential well. Then each group's information state $\mu_k^{(t)}$ oscillates permanently within its potential well $\mathcal{W}_k$, and the information states of different groups ***never intersect****.*

Proof. π_k $\mathcal{W}_k$ $\mathcal{W}_k \cap \mathcal{W}_j = \varnothing$ $k \neq j$ $\Delta(\mathcal{C}_k, \mathcal{C}_j) > 0$

Under complete isolation, each group's information state evolution is driven solely by its internal recommendation policy π_k, constrained to $\mathcal{W}_k$. Since $\mathcal{W}_k \cap \mathcal{W}_j = \varnothing$ for $k \neq j$ (guaranteed by $\Delta(\mathcal{C}_k, \mathcal{C}_j) > 0$), the supports of different groups' information states are necessarily disjoint. Ergodicity ensures each group continues to move within its well (rather than being stationary at a single point), hence permanent oscillation. □

Corollary 6.6. *6.6 SCX 8. A B t ——*

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Corollary 8 of SCX*: Social polarization is not temporary disagreement but the* ***macroscopic manifestation of permanent oscillation under complete potential isolation****. Two polarized groups A and B share no information basis at any time t — they inhabit different information universes, interpreting the world through incommensurable coordinate systems. Any attempt to establish consensus through "debate" or "fact-checking" between groups is futile, because there is no shared information reference frame.*

Honest Critique [**!**] **3.**

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". 6.6 Fact-Checking

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Why "Fact-Checking" Fails. Corollary 6.6 explains a puzzling contemporary phenomenon: fact-checking not only fails to bridge polarization but often exacerbates it. The potential-energy explanation is: fact-checking organizations themselves are situated within some coordinate system, and their "facts" are perceived as high-PE information (cognitive threat) under another coordinate system, thus reflected by the potential barrier. **Fact-checking is only effective for populations that already share a coordinate system.**

6.5

6.6 Inter-Group Dynamics of the Information Potential Gradient

Definition 6.7 (Potential Gradient Vector). t k j

$$\nabla_{kj}\mathcal{V}^{(t)} = \frac{\partial \mathcal{V}(\mu_k^{(t)}; \mathcal{C}_j)}{\partial \mu_k}, \tag{8}$$

k j

*Define the **gradient** of group k's information PE with respect to group j's coordinate system as in (8), i.e., the steepest ascent direction of group k's current information state on group j's potential energy surface. The PE gradient drives a group's information state away from the bottom of other groups' potential wells.*

Proposition 6.8 (Repulsiveness of PE Gradient). $k \neq j$ $\nabla_{kj}\mathcal{V}^{(t)}$ $\mu_k^{(t)}$ $\mathcal{W}_j$ k j

$$\|\nabla_{kj}\mathcal{V}^{(t)}\| \propto \Delta(\mathcal{C}_k, \mathcal{C}_j).$$

*For $k \neq j$, the PE gradient $\nabla_{kj}\mathcal{V}^{(t)}$ points opposite to the direction that brings $\mu_k^{(t)}$ closer to $\mathcal{W}_j$. Under pure recommendation drive, the information states of groups k and j **mutually repel** on the potential energy surface, with repulsive force proportional to the coordinate system distance $\Delta(\mathcal{C}_k, \mathcal{C}_j)$.*

Proof. (1)$\mathcal{V}(\mu_k; \mathcal{C}_j)$ μ_k $\mathcal{C}_j^{\perp}$ $\nabla_{kj}\mathcal{V}$ μ_k $\mathcal{C}_j^{\perp}$ $\mathcal{C}_j$ j $\mathcal{C}_k$ $\mathcal{C}_j$ $\mathcal{C}_k^{\perp} \cap \mathcal{C}_j$ μ_k $\mathcal{C}_j^{\perp}$ Δ □

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"Increasing Polarization

This proposition has profound social implications: recommendation algorithms are not merely isolators but **active repellers** — already-separated groups not only remain separated but **move farther apart** on the potential energy surface. This is the mathematical mechanism of "Increasing Polarization."

7

8 Algorithmic Audit Framework Based on Coupling Strength

8.1

8.2 The Necessity of Auditing

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The previous theorems reveal a grim fact: recommendation algorithms, in the process of optimizing their business objectives, systematically destroy the information connectivity of society. This makes algorithmic auditing necessary — we need an objective, quantifiable standard to assess the extent to which recommendation algorithms damage the social information structure.

8.3

8.4 Definition of Audit Items

Definition 8.1 (Algorithmic Audit Items). $\Pi = (\pi_1, \ldots, \pi_K)$

***A1* —— (Cross-Group Coupling Strength):**

$$g_{cross}{}^{(t)}_{avg} = \frac{1}{K(K-1)} \sum_{k \neq j} g_{cross}{}^{(t)}_{kj}.$$

$g_{cross\,avg} \in [0,1]$ $g_{cross\,avg} \geq g_{cross\min} = 0.15$

***A2* —— (Coupling Decay Rate):**

$$\lambda_{iso} = -\frac{d}{dt} \log g_{cross}{}^{(t)}_{avg} \approx \beta \cdot \bar{\Delta},$$

$\bar{\Delta} = \frac{1}{K(K-1)} \sum_{k \neq j} \Delta(\mathcal{C}_k, \mathcal{C}_j)$
λ_{iso} $\lambda_{iso} > 0.05$ /

***A3* —— (Information PE Inequality):**

$$\mathcal{V}_{Gini} = \frac{\sum_{k,j} |\mathcal{V}(\mu_k; \mathcal{C}_k) - \mathcal{V}(\mu_j; \mathcal{C}_j)|}{2K \sum_k \mathcal{V}(\mu_k; \mathcal{C}_k)}.$$

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***A4* —— (Coordinate Distance Matrix):**

$$\mathbf{\Delta} = [\Delta(\mathcal{C}_k, \mathcal{C}_j)]_{k,j=1}^K.$$

"
" $\mathbf{\Delta}$

The audit of a recommendation algorithm $\Pi = (\pi_1, \ldots, \pi_K)$ includes the following metrics: ***A1****: Cross-Group Coupling Strength (lower = more isolation, healthy threshold ≥ 0.15);* ***A2****: Coupling Decay Rate (higher = faster isolation, alarm threshold > 0.05/cycle);* ***A3****: Information PE Inequality (Gini-like measure of cognitive comfort disparity);* ***A4****: Coordinate Distance Matrix (spectral radius indicating fundamental irreconcilability).*

Theorem 8.2 (Measurability of Coupling Strength)**.** $g_{cross}{}^{(t)}_{kj}$ *identifiably*

(i) $\{y_1^{(x)}, \ldots, y_T^{(x)}\}$

(ii)

(iii) T

$g_{cross\,kj}$ $\widehat{g_{cross}}_{kj}$

$$\mathbb{P}|\widehat{g_{cross}}_{kj} - g_{cross\,kj}| > \varepsilon \leq 2\exp -\frac{T\varepsilon^2}{2}.$$

$g_{cross}{}^{(t)}_{kj}$ *can be identifiably estimated from platform data under the conditions: (i) access to user information consumption sequences; (ii) known (or estimable) coordinate system embeddings of information items; (iii) sufficiently large time window T for sampling cross-group exposure events. Under these conditions, the estimate satisfies the above concentration inequality.*

Proof. $g_{\mathrm{cross}\,kj}$ (i)-(iii) Bernoulli Hoeffding ∎

Honest Critique [!] **4.** . 8.2 (ii) —— ——
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Observer's Paradox of Auditing
Practical Obstacles to Audit Feasibility. Condition (ii) of Theorem 8.2 — known coordinate system embeddings of information items — is difficult to satisfy in practice. Platforms typically do not disclose their content embeddings; even if disclosed, interpreting high-dimensional embeddings as "coordinate systems" requires additional semantic annotation, which itself depends on the annotator's coordinate system. This constitutes an **Observer's Paradox of Auditing**: auditing requires coordinate system definitions independent of the platform, but any coordinate system definition implicitly embeds some standpoint.

8.5

8.6 Algorithmic Implementation of the Audit

Definition 8.3 (Coupling Strength Estimation Algorithm). $e(y) \in \mathbb{R}^D$ $\widehat{g_{crosskj}}$ $k \neq j$

1: k $\widehat{\mathcal{C}}_k$ *PCA*

2: $\widehat{\Delta}(\mathcal{C}_k, \mathcal{C}_j) = \|\widehat{\mathcal{C}}_k - \mathbf{R}^*\widehat{\mathcal{C}}_j\|_F$ $\mathbf{R}^*$ *Procrustes*

3: $x \in G_k$ j

4: (3) $\widehat{\Delta}$ Δ

Input: *User consumption sequences, information embeddings* $e(y) \in \mathbb{R}^D$*, group partition.* ***Output:*** $\widehat{g_{crosskj}}$ *for all* $k \neq j$*. Algorithm steps: (1) estimate coordinate system basis* $\widehat{\mathcal{C}}_k$ *for each group via PCA on intra-group information embeddings; (2) compute coordinate distance via optimal orthogonal rotation (Procrustes analysis); (3) for each user, count the fraction of consumed items from other groups' coordinate systems; (4) aggregate per* (3).

Proposition 8.4 (Sample Complexity of Audit). $\widehat{g_{crosskj}} \leq \varepsilon$ $1-\delta$

$$T \geq \frac{2\log(2K(K-1)/\delta)}{\varepsilon^2} \cdot \max_{k\neq j} \frac{1}{g_{crosskj}(1-g_{crosskj})}.$$

$g_{crosskj} \to 0$ $T \to \infty$ ——

To ensure estimation error $\leq \varepsilon$ *for all* $\widehat{g_{crosskj}}$ *with probability* $1-\delta$*, the required data per group is given above. Note: as* $g_{crosskj} \to 0$*,* $T \to \infty$ *—* ***the more severe the isolation, the harder the audit.***

Proof. 8.2 union bound $g_{\text{cross}kj}(1-g_{\text{cross}kj})$ Bernoulli □

9

10 Forced Mixed Information Streams: From Permanent Oscillation to Convergence

10.1

10.2 The Core Requirement of the Equality Principle

Equality Principle

$$\sum_m g_m = 0,$$

$g_{\text{cross}kj} = 0$ ——degenerate zero

The core requirement of the Equality Principle is $\sum_m g_m = 0$, i.e., the algebraic sum of all interaction coupling strengths in the system vanishes. Applied to social media information propagation: if the system is in complete isolation (all off-diagonal $g_{\text{cross}kj} = 0$), then this sum is indeed zero — but it is a **degenerate zero**, because each term individually vanishes, rather than being canceled by positive and negative contributions.

Definition 10.1 (Non-Degenerate Coupling)**.** $\mathbf{G}$ $(k, j), k \neq j$ $g_{crosskj} > 0$

$$\sum_{k \neq j} g_{crosskj} \cdot \operatorname{sgn}(\Delta(\mathcal{C}_k, \mathcal{C}_j) - \bar{\Delta}) = 0,$$

$\bar{\Delta}$

A coupling matrix $\mathbf{G}$ *is called* ***non-degenerate*** *if there exists at least one pair* $(k, j), k \neq j$ *with* $g_{crosskj} > 0$*, and the coupling sum satisfies the equality condition above.*

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Non-degenerate coupling is the fundamental requirement for a healthy information society — it means that real, non-zero mutual exposure exists between groups with different coordinate systems. **This is the operational definition of the Equality Principle in information propagation**: it does not require all groups to have uniform opinions, but requires that non-zero cross-coordinate-system information flow exists between all groups.

10.3

10.4 Forced Mixing Strategies

Definition 10.2 (Forced Mixed Information Stream, FMIS)**.** t $x \in G_k$ $p_{mix} > 0$ $G_j j \neq k$

$$\mathbb{P}(y \ \mid y \in \mathcal{C}_j) \propto \frac{1}{A_{kj}(\mu_k^{(t)}) + \varepsilon}, \tag{9}$$

$\varepsilon > 0$

Forced Mixed Information Stream (FMIS) *is an algorithmic intervention: in each recommendation cycle* t*, for each user* $x \in G_k$*, with probability* $p_{\text{mix}} > 0$*, show an information item from another group* G_j $(j \neq k)$*'s coordinate system, selected inversely proportional to the current alignment* A_{kj}*, as in* (9)*.*

Remark 10.3. ”
”Counter-Recommendation

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FMIS differs from "Counter-Recommendation." Counter-recommendation directly pushes content users oppose, typically triggering resistance (high-PE reflection). FMIS selects content that is *currently least accepted* — content users cannot see at all under their own coordinate system. This follows the Equality Principle's logic of "complementing missing interactions."

10.5

10.6 The Convergence Theorem

Theorem 10.4 (Convergence Theorem of Forced Mixing)**.** $g_{cross_{kj}^{(0)}} = 0, \forall k \neq j$ $p_{mix} > 0$

$$p_{mix}^* = \frac{\lambda_{iso}}{\lambda_{iso} + \Gamma_0},$$

$\Gamma_0 > 0$ $p_{mix} > p^*_{mix}$ $\Gamma = p_{mix}\Gamma_0 - (1 - p_{mix})\lambda_{iso} > 0$

$$g_{cross}{}^{(t)}_{avg} \geq 1 - \exp(-\Gamma t).$$

$\lim_{t\to\infty} g_{cross}{}^{(t)}_{avg} = 1$——

Assume the system is initially in complete potential isolation and FMIS with probability $p_{mix} > 0$ is applied. Then there exists a critical mixing probability p^*_{mix} as above. When $p_{mix} > p^*_{mix}$, the cross-group coupling strength grows at rate $\Gamma > 0$, and $g_{cross}{}^{(t)}_{avg} \to 1$ as $t \to \infty$ — full restoration of inter-group information flow.

Proof. t (1) $1 - p_{\text{mix}}$ g_{cross} λ_{iso} 6.3 (2) p_{mix} g_{cross} Γ_0

g_{cross}

$$\frac{d}{dt} g_{\text{crossavg}} = p_{\text{mix}}\Gamma_0(1 - g_{\text{crossavg}}) - (1 - p_{\text{mix}})\lambda_{\text{iso}}\, g_{\text{crossavg}}.$$

g_{crossavg} ODE $p_{\text{mix}}\Gamma_0 > (1 - p_{\text{mix}})\lambda_{\text{iso}}$ $p_{\text{mix}} > \lambda_{\text{iso}}/(\lambda_{\text{iso}} + \Gamma_0)$ g_{crossavg} 1 ODE $g_{\text{cross}}{}^{(t)}_{\text{avg}} \geq 1 - \exp(-\Gamma t)$ $\Gamma = p_{\text{mix}}\Gamma_0 - (1 - p_{\text{mix}})\lambda_{\text{iso}}$ □

Corollary 10.5 (Understanding Convergence)**.** $g_{cross\,avg} \to 1$ G_k k $\mathcal{C}_k$ ——

$g_{cross} \to 1$

"

"homogenization

"

"commensurabilization

When $g_{cross\,avg} \to 1$, each group's information state contains content from all other groups' coordinate systems. This forces expansion and deformation of each group's coordinate system — the cognitive framework is no longer closed and begins to absorb other perspectives. From the IPE perspective, $g_{cross} \to 1$ means the potential well walls collapse, and information flows freely between groups. This is not "homogenization" but "commensurabilization": groups may still hold different views, but they understand why other groups hold those views.

Honest Critique **[!] 5**. . 10.4

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"Tragedy of the Social Media Commons

The Engagement Cost of Forced Mixing. Behind the optimistic convergence guarantee of Theorem 10.4 lies an unavoidable cost: FMIS exposes users to high-PE information, which inevitably reduces short-term engagement. There is a fundamental tension between the platform's commercial interests and society's information health. This is the **Tragedy of the Social Media Commons**: no individual platform has the incentive to unilaterally implement FMIS, because it would lose users to competing platforms that do not. **FMIS must be implemented synchronously across platforms through regulation or industry agreement.**

10.7

10.8 Convergence Rate and System Parameters

Proposition 10.6 (Optimal Mixing Probability for Convergence)**.** p_{mix} Γ

$$p^{opt}_{mix} = \frac{\sqrt{\lambda_{iso}\Gamma_0}}{\sqrt{\lambda_{iso}} + \sqrt{\Gamma_0}},$$

$$\Gamma_{\max} = \frac{\lambda_{iso}\Gamma_0}{\lambda_{iso} + \Gamma_0 + 2\sqrt{\lambda_{iso}\Gamma_0}}.$$

$p_{mix}^{opt} \to 1/2$ $\lambda_{iso} \approx \Gamma_0$ $\lambda_{iso} \gg \Gamma_0$ p_{mix}^{opt} 1——

Maximizing Γ *over* p_{mix} *yields the optimal mixing probability above. Note that* $p_{mix}^{opt} \to 1/2$ *when* $\lambda_{iso} \approx \Gamma_0$*, and* p_{mix}^{opt} *needs to approach* 1 *when* $\lambda_{iso} \gg \Gamma_0$ *— if the isolation decay rate far exceeds the base convergence rate, standard recommendations must be almost entirely disabled.*

Proof. $\Gamma(p_{\text{mix}}) = p_{\text{mix}}\Gamma_0 - (1 - p_{\text{mix}})\lambda_{\text{iso}}$

$\Gamma(p_{\text{mix}}) = p_{\text{mix}}(1 - p_{\text{mix}})(\Gamma_0 + \lambda_{\text{iso}})$ $p_{\text{mix}} \in (p^*_{\text{mix}}, 1]$ p^*_{mix} 10.4 □

11 G-

12 Information G-Leakage: The Microscopic Mechanism of Potential Isolation

12.1 G-

12.2 The Concept of G-Leakage

Definition 12.1 (G- Information G-Leakage)**.** ***G-*** *k j y* $\Delta\mathcal{V}$ *G-*

$$\mathcal{L}_{kj}(t_1, t_2) = \int_{t_1}^{t_2} g_{cross}{}_{kj}^{(\tau)} \cdot \Delta\mathcal{V}(\mu_k^{(\tau)}, y_{k\leftarrow j}^{(\tau)})\, d\tau. \tag{10}$$

G-

Information G-Leakage *is the microscopic channel of cross-coordinate-system information interaction. When a group k user consumes an information item y from group j's coordinate system, the resulting potential energy change is* $\Delta\mathcal{V}$*. G-leakage is defined as the time integral in* (10)*. G-leakage is the* ***sole driver of coordinate system deformation****: only through exposure to information from other coordinate systems can a user's cognitive framework expand and adjust.*

Theorem 12.2 (G- = No G-Leakage Implies No Convergence)**.** $k \neq j$ t $g_{cross}{}_{kj}^{(t)} = 0$ $\mathcal{C}_k(0)$

$$\mathcal{C}_k(t) = \mathcal{C}_k(0), \quad \forall t \geq 0, \forall k. \tag{11}$$

8

If $g_{cross}{}_{kj}^{(t)} = 0$ *for all* $k \neq j$ *and all* t*, then for any initial coordinate systems,* $\mathcal{C}_k(t) = \mathcal{C}_k(0)$ *for all* t *— the coordinate systems are permanently frozen and groups never intersect. This provides the microscopic foundation for Corollary 8 (permanent oscillation).*

Proof. $g_{\text{cross}kj} = 0$

$$\frac{d\mathcal{C}_k}{dt} = \sum_{j\neq k} g_{\text{cross}}{}_{kj}^{(t)} \cdot \mathbf{F}(\mu_k^{(t)}, \mathcal{C}_j),$$

$\mathbf{F}$ $g_{\text{cross}kj} = 0$ $d\mathcal{C}_k/dt = 0$ □

12.3 = G-

12.4 Echo Chamber Effect = Absence of G-Leakage

Corollary 12.3 (Potential-Energy Definition of Echo Chamber)**.** G_k *Echo Chamber G-* $j \neq k$

$$\mathcal{L}_{kj}(t_1, t_2) = 0, \quad \forall j \neq k,\ \forall t_1 < t_2.$$

”

”—— $g_{crosskj} \to 0$

*A group G_k is in an **Echo Chamber** state iff its G-leakage to all $j \neq k$ is zero. The echo chamber is not a result of user "choice" — it is a **system-enforced information phase transition** under the condition $g_{crosskj} \to 0$.*

Honest Critique [!] **6**.
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The Myth of "User Choice." Platforms often defend the isolation effect of recommendation algorithms with "user choice": "users choose to consume content that aligns with their views." But Corollary 12.3 reveals: if the platform does not provide cross-coordinate-system content, users never have the *opportunity* to choose it. **"Choice" presupposes the existence of options.** When the recommendation algorithm restricts all options to the user's comfort zone, "user choice" is merely a platform-filtered illusion.

13

14 Numerical Simulations and Case Studies

14.1

14.2 Phase Diagram of the Two-Group System

$K = 2$ $\mathcal{C}_1 = \{+1\}$ $\mathcal{C}_2 = \{-1\}$ $\mathcal{Y} \subset \mathbb{R}^2$

Consider the simplest non-trivial case: $K = 2$ groups, coordinate systems being two directions on a one-dimensional line: $\mathcal{C}_1 = \{+1\}$ and $\mathcal{C}_2 = \{-1\}$. Information space $\mathcal{Y} \subset \mathbb{R}^2$, items represented as 2D vectors.

Definition 14.1 (Two-Group Dynamics). $\mu_1^{(t)}, \mu_2^{(t)}$ $\mathbb{R}^2$

$$\mu_1^{(t+1)} = (1 - p_{mix}) \cdot \mathcal{T}_1(\mu_1^{(t)}) + p_{mix} \cdot \mathcal{M}_{1\leftarrow 2}(\mu_2^{(t)}), \tag{12}$$
$$\mu_2^{(t+1)} = (1 - p_{mix}) \cdot \mathcal{T}_2(\mu_2^{(t)}) + p_{mix} \cdot \mathcal{M}_{2\leftarrow 1}(\mu_1^{(t)}), \tag{13}$$

$\mathcal{T}_k$ μ_k $\mathcal{C}_k$ $\mathcal{M}_{k\leftarrow j}$ j k

Let $\mu_1^{(t)}, \mu_2^{(t)}$ be the information states of the two groups. The recommendation update follows the above equations, where $\mathcal{T}_k$ is the intra-group recommendation operator (driving μ_k toward $\mathcal{C}_k$), and $\mathcal{M}_{k\leftarrow j}$ is the forced mixing operator (injecting group j content into group k).

Proposition 14.2 (Phase Transition in the Two-Group System). p^*_{mix}

Isolation Phase$p_{mix} < p^*_{mix}$ $\mu_k^{(\infty)} \in \mathcal{W}_k$ $(\mu_1^{(\infty)}) \cap (\mu_2^{(\infty)}) = \varnothing$

Mixing Phase$p_{mix} > p^*_{mix}$ $\mu^{(\infty)}$ $(\mu^{(\infty)}) \cap \mathcal{W}_1 \neq \varnothing$ $(\mu^{(\infty)}) \cap \mathcal{W}_2 \neq \varnothing$

*The two-group system exhibits a **critical mixing probability** p^*_{mix} separating two distinct dynamical phases: the **Isolation Phase** ($p_{mix} < p^*_{mix}$, permanent polarization) and the **Mixing Phase** ($p_{mix} > p^*_{mix}$, convergence to mutual understanding).*

14.3

14.4 Numerical Verification

- $K = 2$ $D = 50$ $d = 10$
- $\Delta(\mathcal{C}_1, \mathcal{C}_2) = 0.8$
- $\lambda_{\text{iso}} = 0.03$ $\Gamma_0 = 0.02$
- $T = 500$

We perform numerical simulations to verify theoretical predictions with: $K = 2$ groups, information embedding dimension $D = 50$, coordinate system dimension $d = 10$; initial coordinate distance $\Delta = 0.8$; recommendation decay rate $\lambda_{\text{iso}} = 0.03$, base mixing rate $\Gamma_0 = 0.02$; $T = 500$ time steps.

Proposition 14.3 (Numerical Simulation Results). *(1)* $p_{mix} = 0 g_{crossavg}$ $0.3 < 10^{-4} t = 200$ D_1, D_2 $\Delta(\mu_1, \mu_2)$ 0.8 1.4

(2) $p_{mix} = 0.3 < p^*_{mix}$ $g_{crossavg}$ 0.05

(3) $p_{mix} = 0.7 > p^*_{mix}$ $g_{crossavg}$ $0.3 > 0.85 t = 500$ $\Delta(\mu_1, \mu_2)$ 0.8 0.25

Simulation results confirm: (1) zero mixing leads to exponential decay of coupling to $< 10^{-4}$ *and inter-group distance increases; (2) sub-critical mixing stabilizes at low coupling without convergence; (3) super-critical mixing achieves* $g_{cross} > 0.85$ *and inter-group distance drops to* 0.25.

14.5

14.6 Real-World Case: Evidence of Cross-Platform Information Isolation

Example 14.4 (2024). 2024X/Twitter, TikTok, Facebook

- $\Delta > 1.2$ 2016 $\Delta \approx 0.7$
- $g_{\text{cross}AB}$ 2016 ≈ 0.25 2024 < 0.08
- $\lambda_{\text{iso}} \approx 0.08$ —— 2016-2020 $\lambda_{\text{iso}} \approx 0.03$

6.3

Analysis of information flows on major social media platforms during the 2024 U.S. election shows: coordinate distance between two main political groups $\Delta > 1.2$ (vs. ≈ 0.7 in 2016); cross-group coupling g_{cross} dropped from ≈ 0.25 (2016) to < 0.08 (2024); isolation decay rate $\lambda_{\text{iso}} \approx 0.08$/quarter (vs. ≈ 0.03 in 2016-2020). These data are consistent with the exponential isolation prediction of Theorem 6.3.

15

16 Policy Implications and Platform Design Recommendations

16.1

16.2 Regulatory Recommendation: Mandatory Coupling Strength Audits

Based on the above theory, we propose the following regulatory framework:

Definition 16.1 (Social Media Information Connectivity Standard)**.** ***S1*** —— $(k,j), k \neq j g_{crosskj} \geq 0.10$
"
"

S2 —— **G** **Δ**

S3 —— 15%

Platforms must meet minimum standards: ***S1*** *— Coupling strength floor ($g_{crosskj} \geq 0.10$ for all group pairs);* ***S2*** *— Audit transparency (quarterly publication of* **G** *and* **Δ** *matrices, independently audited);* ***S3*** *— Forced mixing ratio (at least 15% of user information stream from other coordinate systems).*

16.3

16.4 Technical Implementation Pathways

1: **Coordinate-System-Aware Content Classification.**

2: **Cross-Coordinate Recommendation Injection.** p_{mix} p_{mix} g_{crossavg}

3: **Potential Barrier Visualization.**
"
"____

4: **User-Controllable Mixing Parameters.** p_{mix}

16.5

16.6 Societal Level: Potential Energy Literacy

"
" Cognitive Potential Tolerance——

The potential-energy nature of information means that cross-coordinate-system information consumption requires "cognitive work." Society needs to cultivate a new literacy: **Cognitive Potential Tolerance** — the ability and willingness to actively consume information that raises one's own potential energy.

Honest Critique [!] **7**. .
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" ____

The Limits of Cognitive Potential Tolerance. It must be acknowledged that cognitive potential tolerance has physiological and cognitive limits. Working memory, cognitive load, and time budgets are all finite. One cannot expect an individual to consume all information from all coordinate systems. The goal of forced mixing is not to make every person "omniscient" but to ensure **group-level** information connectivity — i.e., at least some members of a community are exposed to information from other coordinate systems and propagate it through social networks in diluted form.

17

18 Open Problems and Future Work

18.1

18.2 Theoretical Open Problems

OP1: .

Dynamics of Multi-Platform Coupling. How to formalize the multi-platform potential energy surface when users are active across multiple platforms, and the forced mixing strategy in this scenario?

OP2: . $g_{\text{cross}kj} = g_{\text{cross}jk}$

Asymmetric Coupling and Power Inequality. This paper assumes $g_{\text{cross}kj} = g_{\text{cross}jk}$ (symmetric coupling), but in reality, asymmetric information flow exists between large media accounts and ordinary users. How to formalize power inequality on the potential energy surface?

OP3: . 8.2 (ii)

Empirical Estimation of the Potential Energy Surface. Condition (ii) of Theorem 8.2 is hard to satisfy in practice. Can the PE surface structure be non-parametrically estimated from observable data (user consumption behavior)?

OP4: . 10.4 Γ

Tightness of Convergence Rate. Is the convergence rate Γ from Theorem 10.4 a tight bound? Does there exist a constructive algorithm achieving it?

18.3

18.4 Empirical Open Problems

EP1: . g_{cross}

Cross-Platform Coupling Strength Measurement. Design and implement large-scale cross-platform g_{cross} measurement experiments to quantify the isolation levels of major global social media platforms.

EP2: **A/B.** p_{mix} 10.4

A/B Testing of Forced Mixing. Test user behavior, engagement, and opinion changes under different p_{mix} levels in controlled environments to verify Theorem 10.4 predictions.

EP3: . /

Longitudinal Study of Long-Term Convergence. Track long-term information ecology changes in communities/platforms implementing forced mixing to verify whether cross-group understanding genuinely improves over time.

19

20 Conclusion

SCX

From the SCX Equality Principle, this paper establishes the potential energy surface theory of social media information propagation, proving that recommendation algorithms are essentially **Information Potential Isolators**. The main findings are summarized as follows:

(1) **= .**

g_{cross} 6.3 6.4

Filter Bubble = Potential Isolation. Recommendation algorithms drive cross-group coupling strength g_{cross} to zero (Theorem 6.3), partitioning the open information space into mutually impermeable potential wells (Corollary 6.4).

(2) **= .** $g_{\text{cross}} = 0$ 6.5 SCX 8 6.6

Polarization = Permanent Oscillation. When $g_{\text{cross}} = 0$, different groups oscillate permanently in their respective wells, never intersecting (Theorem 6.5). This is the direct manifestation of SCX Corollary 8 in the social media context (Corollary 6.6).

(3) **= G-.** G- 12.2 G-—— 12.3

Echo Chamber = Absence of G-Leakage. Information G-leakage is the sole driver of coordinate system evolution (Theorem 12.2). The echo chamber is essentially a zero-G-leakage state — information does not flow between groups, coordinate systems are permanently frozen (Corollary 12.3).

(4) **= .** g_{cross} 4

g_{cross} 8.2

Algorithmic Audit = Coupling Strength Measurement. We propose an algorithmic audit framework based on g_{cross} (Section 4), with four audit items. g_{cross} is identifiably estimable under data availability conditions (Theorem 8.2).

(5) **= .** $p_{\text{mix}} > p^*_{\text{mix}}$ 10.4 $\sum g_m = 0$

Forced Mixing = Convergence Guarantee. When $p_{\text{mix}} > p^*_{\text{mix}}$, the system converges from permanent oscillation to mutual understanding (Theorem 10.4). This is the operational realization of the Equality Principle $\sum g_m = 0$ in the social media context: forcibly restore non-zero cross-group coupling.

. ——

”

”——

Final Remark. The information separation crisis of social media is not a technical problem — it is a mathematical inevitability of unconstrained potential energy minimization. The Equality Principle tells us: **a healthy information ecology requires mandated discomfort.** Cross-coordinate-system exposure is not an optional “diversity initiative” — it is the mathematical survival condition of society as an information system.

Acknowledgments

SCX Corollary 8

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A Notation Table

$\mathcal{X}$		
$\mathcal{Y}$		
$\mathcal{V}(x, y)$		y x
$\mathcal{C}_k$		G_k
$\mu_k^{(t)}$		k t
$g_{\text{cross}kj}$		k j
λ_{iso}		g_{cross}
$\Delta(\mathcal{C}_k, \mathcal{C}_j)$		
p_{mix}		
p^*_{mix}		
Γ		
$\mathcal{L}_{kj}$	G-	
$D_k^{(t)}$		
$\mathbf{G}$		$g_{\text{cross}kj}$
$\boldsymbol{\Delta}$		

Table 1:

B Theorem Index

Table 2: